66. A number consists of three digits of which the middle one is zero and their sum is 4. If the number formed by interchanging the first and last digits is greater than the number itself by 198, then the difference between the first and last digits is

(a) 1 (b) 2
(c) 3 (d) 4

Ans. (b) Let unit digit of the number = x
and hundred digit of the number = y
according to question,
$\because$ the number = 100y + x
and number after interchanging =100x + y
then (100x + y) – (100y + x) =198
100x + y – 100y –x = 198
99x – 99y = 198
$\boxed{x - y = 2}$

67. A solid cube of 3 cm side, painted on all its faces, is cut up into small cubes of 1 cm side. How many of the small cubes will have exactly two painted faces?

(a) 12 (b) 8
(c) 6 (d) 4

Ans. (a) The number of two painted
Face = 12 (x – 2)
= 12 (3 – 2)
= 12 × 1
= 12

68. While writing all the numbers from 700 to 1000, how many numbers occur in which the digit at hundred's place is greater than the digit at ten's place, and the digit at ten's place is greater than the digit at unit's place?

(a) 61 (b) 64
(c) 85 (d) 91

Ans. (c)
For required numbers,
From 700 to 800,
1 + 2 + 3 + 4 + 5 + 6 = 21
From 800 to 900,
1 + 2 + 3 + 4 + 5 + 6 + 7 = 28
From 900 to 1000,
1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 = 36
$\because$ Total numbers = 21 + 28 + 36 = 85

69. A bookseller sold 'a' number of Geography textbooks at the rate of ₹ x per book, a+2 number of History textbooks at the rate of ₹ (x+2) per book and 'a–2' number of Mathematics textbooks at the rate of ₹ (x-2) per book. What is his total sale in ₹ ?

(a) $3x + 3a$ (b) $3ax + 8$
(c) $9ax$ (d) x^3a^3

Ans. (b)
Total sell = ax + (a + 2) (x + 2)
+ (a – 2) (x –2)
= ax + ax + 2x + 2a + y
+ ax – 2x – 2a + y
$\Rightarrow 3ax + 8$

70. A bag contains 15 red balls and 20 black balls. Each ball is numbered either 1 or 2 or 3. 20% of the red balls are numbered 1 and 40% of them are numbered 3. Similarly, among the black balls, 45% are numbered 2 and 30% are numbered 3. A boy picks a ball at random. He wins if the ball is red and numbered 3 or if it is black and numbered 1 or 2. What are the chances of his winning?

(a) $\frac{1}{2}$ (b) $\frac{4}{7}$
(c) $\frac{5}{9}$ (d) $\frac{12}{13}$

Ans. (b)
$\because$ The number of red balls = 15
The number of red balls which are numbered 1
$\Rightarrow 15 \times \frac{20}{100} \Rightarrow 3$
The number of red balls which numbered 3
$= 15 \times \frac{40}{100} = 6$
The number of red balls which are numbered 2
=15 – (6 + 3) = 6
$\because$ The number of black balls = 20
The number of which are numbered to = $20 \times \frac{45}{100} = 9$
The number of black balls which are numbered 3
$= 20 \times \frac{30}{100} = 6$
the number of black balls which are numbered 1 = 20 – (9 + 6) = 5
The probability of winning
$\Rightarrow$ The number of Red balls which are numbered 3
+ the number of black balls which are
+ numbered 1 + the number of black ball which are numbered 2
$\Rightarrow$ total number of black and Red balls
$= \frac{6+5+9}{15+20}$
$= \frac{20}{35} = \frac{4}{7}$

71. A student has to get 40% marks to pass in an examination. Suppose he gets 30 marks and fails by 30 marks, then what are the maximum marks in the examination?

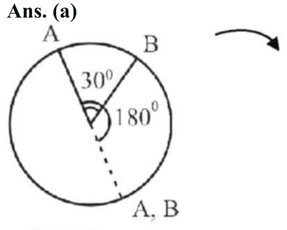
(a) 100 (b) 120
(c) 150 (d) 300

Ans. (c)
Let maximum number = m
according to questions,
$m \times \frac{40}{100} = 30 + 30$
$m \times \frac{40}{100} = 60$
$m \times \frac{2}{5} = 60$
$\boxed{m = 150}$

72. Two persons, A and B are running on a circular track. At the start, B is ahead of A and their positions make an angle of 30^0 at the centre of the circle. When A reaches the point diametrically opposite to his starting point, he meets B. What is the ratio of speeds of A and B, if they are running with uniform speeds?

(a) 6 : 5 (b) 4 : 3
(c) 6 : 1 (d) 4 : 2

Ans. (a)



$\because \boxed{V = \frac{d}{t}}$
$d_A = \frac{180}{360} \times 2\pi r$
$d_B = \frac{(180-30)}{360} \times 2\pi r$
$\because V_A : V_B = \frac{d_A}{t_A} : \frac{d_B}{t_B}$
$V_A : V_B = \frac{\frac{180}{360} \times 2\pi r}{t} : \frac{\frac{(180-30)}{360} \times 2\pi r}{t}$
$\boxed{V_A : V_B :180 : 150}$
$\boxed{V_A : V_B = 6 : 5}$ [$\because$taken time is equal]

73. A shopkeeper sells an articale at ₹ 40 and gets x% profit. However, when he sells it at ₹ 20, he faces same percentage of loss. What is the original cost of the article?

(a) ₹10 (b) ₹20
(c) ₹30 (d) ₹40

Ans. (c) $\because$ P =SP – CP
and L= CP – SP
According to question,
P = L
SP – CP = CP – SP
40 – x = x – 20
2x = 60
$\boxed{x = 30₹}$

74. There are 24 equally spaced points lying on the circumference of a circle. What is the maximum number of equilateral triangles that can be drawn by taking sets of three points as the vertices?

(a) 4 (b) 6
(c) 8 (d) 12

Ans. (c)
The maximum number of triangles $= \frac{24}{3} = 8$

75. Consider the sequence given below : 4/12/95, 1/1/96, 29/1/96, 26/2/96,..... What is the next term of the series?

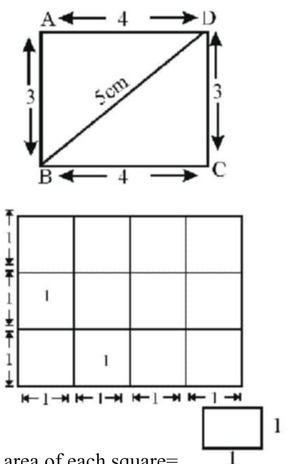
(a) 24/3/96 (b) 25/3/96
(c) 26/3/96 (d) 27/3/96

Ans. (b)
$4/12/95 \xrightarrow{+28\text{days}} 01/01/1996 \xrightarrow{+28\text{days}}$
$29/01/96 \xrightarrow{+28\text{days}} 26/02/96$
$\xrightarrow{+28\text{days}} 25/03/96$

76. Twelve equal squares are placed to fit in a rectangle of diagonal 5 cm. There are three rows containing four squares each. No gaps are left between adjacent squares. What is the area of each square?

(a) $\frac{5}{7}$ sq cm (b) $\frac{7}{5}$ sq cm
(c) 1 sq cm (d) $\frac{25}{12}$ sq cm

Ans. (c)



The area of each square=
$= 1 \times 1 = 1 \text{ cm}^2$